Jiaming Chu

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Education

present – Sep 2025 | School of Automation and Intelligent Sensing, **Shanghai Jiao Tong University**
M.Eng. in Electronic Information

Jul 2025 – Sep 2021 | School of Instrumentation Science and Engineering, **Harbin Institute of Technology**
B.Eng. in IntelliSense Engineering

Research

2026 | **VGGT-Splat: Feed-forward VGGT Poses vs. COLMAP for 3D Gaussian Splatting** page • code

- Controlled experiment with **72 runs** on 6 scenes × 8 / 16 / 32 / all frames.
- With few frames COLMAP fails on most scenes while VGGT never does; with all frames COLMAP leads by **2.1–4.6 dB** on five of six scenes, likely because VGGT's poses are less precise, not because it covers less.
- A VGGT depth prior reshapes geometry but barely changes novel views: the few-frame bottleneck is unseen regions.

2026 | **PoseGuard-NeRF: When Does Joint Camera-Pose Optimization Help NeRF?** page • code

- Controlled experiment with **45 vanilla-NeRF runs** (9 pose-noise levels × 5 methods) on when joint pose optimization helps or hurts.
- A camera learning-rate warm-up recovers much of the loss from noisy poses: 14.1 → **22.7 dB** PSNR at 5% translation noise.
- Naive pose optimization from perfect poses drops PSNR 27.1 → 17.9 dB, yet a Sim(3) fit on training cameras recovers **+5.5 dB**: mostly global drift, not a worse reconstruction.
- Released v1.0.0 with reproducible configs, GitHub Actions CI and uv environments.

2026 | **AFVE-Bench: Abandoned-Frame Verbalization in Multi-Turn LLM Revision** preprint

- Benchmark for revisions that adopt the user's new frame yet still expose the abandoned one: **192** bilingual dialogues, **4,204** doubly annotated responses from 11 model configurations.
- The error hits **25.5%** of history-conditioned responses; **19.8%** of target-passing ones still verbalize the old frame.

Experience

Jul 2023 – Jan 2023 | **Captain, Team Lilac 12** @ HIT Smartcar & Innovation Club

- Led vision support; built ROS packages, tuned navigation (SLAM, path planning) and wrote docs.

Apr 2024 – Nov 2023 | **Team Member** @ HIT Center of Ultra-Precision Optoelectronic Instrument Engineering

- Single-beam 3-DoF ultra-precision laser interferometer: host communication and data visualization; the entry won a national second prize (see Awards).

Jan 2024 – Dec 2023 | **Mathematical Modeling (Team Lead)**

- Built the models, wrote the code and the paper; won an ICM Honorable Mention.

Other Projects

Undergrad

- Qingzhou robot (C/C++, Python, Jetson Nano, STM32): SLAM, navigation, visual recognition.
- Vision-controlled car (Python, Raspberry Pi): drives by recognizing QR codes and hand gestures.
- Smart-home system (Python, Raspberry Pi): real-time sensing, logging, display and RF device control.

Honors & Awards

2024	**Honorable Mention**, Interdisciplinary Contest in Modeling (ICM), COMAP
2023	**Second National Prize**, 7th China (International) Sensor Innovation Contest
2023	**Provincial Second Prize**, National University Students Intelligent Car Race

Skills & Languages

Programming	Python, C/C++, MATLAB
Frameworks	PyTorch, OpenCV, NumPy
Systems	Linux, ROS
Dev Tools	Git, Conda, uv, LaTeX
Languages	Chinese (native), English (proficient)